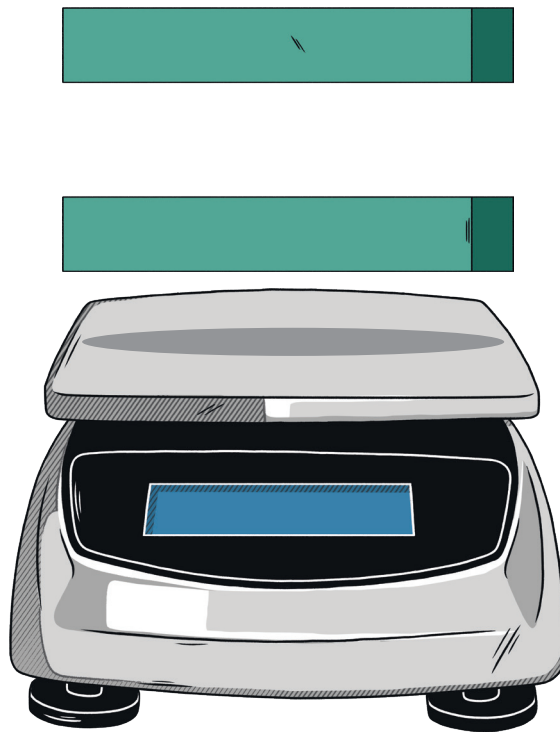


Solving Equations

An equation is a mathematical statement that is made up of two algebraic expressions separated by an equals sign. The algebraic expressions can contain variables, coefficients and constants.

You can think of the equals sign as meaning '**the same as**'. This will help you imagine that either side of the equals sign should always **balance**.



That is, if you do something to one side, you **must** also do it to the other.

Considering the inverse can help you decide what to do next.

For example, in this equation the opposite of adding 5 is to subtract 5:

$$x + 5 = 7$$

$$\begin{array}{r} -5 \quad -5 \\ x + 5 = 7 \end{array}$$

$$x = 2$$

Don't forget to apply **BIDMAS**

Tricky brackets? Try **ex(pand)(ing)** them first.

And finally, remember you can always double check your answer by substituting it back into the original equation.